Completing the numerical intransitive-side support bound

Human Compression project: bounded L044 proof component

Throughout, $G = \langle \rho_0, \ldots, \rho_{r-1} \rangle \leq A_n$ is transitive, $n \geq 12$, its marking is a connected string C-marking, and every maximal omission is intransitive. The unique eligible interior i-edge $\{a, b\}$ joins the two G_i-orbits A, B. Assume explicitly that $|B| \geq 2$ and that $L = G_{<i}$ fixes B pointwise and is intransitive on A. Put $R = G_{>i}$ and $\tau = \rho_{i+1}$. Write m for the number of L-orbits in A and k for their common size.

We use the already proved faithful commuting-orbit facts: L is faithful on each such orbit, $k \geq i + 1$, and τ exchanges $\beta = a^L$ with a distinct orbit β'. Every generator of index at least $i+2$ fixes β pointwise, and those of index at least $i+3$ fix β' pointwise. The only generator of G_i which can move b is τ; since B is nontrivial, it does. These facts use the unique-crossing lemma, commuting factors and Whiston's independent-length bound, not IP of a projected action.

Lemma 1 (Distance to a moved block)**.** *If the last generator has nontrivial image on the L-orbits in A, and $d = r - 1 - i \geq 2$, then $m \geq d + 1$.*

Proof. In the block graph, take a shortest path from β, using indices $i + 1, \ldots, r - 2$, to a block moved by ρ_{r-1}. Such a block is reachable: otherwise the set of reachable blocks would be invariant under every generator of R, hence would be all blocks by transitivity, contradicting the assumed nontrivial last block image. Every intermediate index occurs in this path word. If index j were missing, the factors below and above j could be reordered, since they commute. The upper factors fix β; the lower factors commute with ρ_{r-1}, which fixes β as well. The endpoint block would then be fixed by ρ_{r-1}, a contradiction. The path therefore has at least $d - 1$ edges and d distinct blocks. All earlier blocks are fixed by the last generator, by minimality. The image of the last block under that generator is a further block, proving the bound. □

Proposition 2 (A final generator supported on the first orbit)**.** *If ρ_{r-1} fixes B pointwise, then $r \leq \lfloor (n - 1)/2 \rfloor$.*

Proof. The last generator is nonidentity and supported on A. Its block image cannot belong to the group generated by the other R block images. If it did, those other images would remain transitive on the blocks. Together with L they would make the global last omission transitive on A. Adjoining a generator supported on that one omission orbit could not join any omission orbits. Since G is transitive, the omission would already be transitive, a contradiction. In particular the last block image is nontrivial. Also $d = r - 1 - i \geq 2$, because τ moves B. Lemma 1 gives

$$n = km + |B| \geq (i + 1)(d + 1) + 2 = 2r + (i - 1)(d - 1).$$

For $i \geq 2$ this is at least $2r + 1$, as required.

Suppose $i = 1$ and the rank bound fails. All the inequalities must then be equalities:

$$k = 2, \qquad |B| = 2, \qquad m = d + 1, \qquad n = 2r = 2m + 2.$$

The generator ρ_0 exchanges every one of the m pairs in A and fixes B. It is even, so m is even; $n \geq 12$ gives $m \geq 6$.

Every distinguished R block image is essential. The image of τ is essential because all the others fix β. Each other generator fixes b, hence all of the two-point set B. If its block image

were dispensable, the same global omission argument as above would apply to that generator. Thus the $d = m - 1$ block images are an independent involutory string in S_m. Whiston's equality theorem makes the image S_m. The paid maximal symmetric-string classification gives the ordinary adjacent-transposition path, except possibly the outer degree-six Coxeter action. The latter has transitive endpoint omissions, whereas omitting τ here fixes β, so it is excluded. This uses the accepted complete degree-six certificate and the uniform theorem for $m \geq 7$; it assumes no projected IP.

Number the pairs $0, \ldots, m-1$ so the successive R block images swap $0, 1$, then $1, 2$, and so on. The anchor pair is pair 0: it is the fixed point of the block subgroup omitting τ. Put $H = \langle \rho_{i+3}, \ldots, \rho_{r-1} \rangle$. This group fixes the first two pairs pointwise and induces S_{m-2} on the remaining $m-2$ pairs. On the first four points, τ exchanges two pairs, so is a product of two transpositions. On the remaining points it centralizes H. The accepted paired-row centralizer lemma applies because $m-2$ is even and at least four: any such centralizing involution is even. Hence τ is even on A but is a single transposition on B. This contradicts $G \leq A_n$. □

Proposition 3 (The last generator shared by both orbits)**.** *If ρ_{r-1} acts nontrivially on both A and B, then $r \leq \lfloor (n-1)/2 \rfloor$.*

Proof. Put $d = r - 1 - i$. For $d \geq 2$, the accepted active-orbit distance lemma gives $|A| \geq kd$. The fixed-prefix/no-gap lemma, applied in B and including the final edge, gives a simple path with at least $d+1$ vertices there. Consequently

$$n \geq d(k+1) + 1 \geq d(i+2) + 1 = 2r + i(d-2) - 1. \tag{1}$$

If $i \geq 2$ and $d \geq 3$, this is at least $2r+1$. If $i = 1$ and $d \geq 4$, it is again at least $2r+1$. The remaining case $i = 1, d = 3$ has $r = 5$, and the bound follows directly from $n \geq 12$.

For $d = 2$, the first orbit on which the last generator can act is β'. If it moves β' to another orbit, the active-orbit lemma gives $|A| \geq 3k$, while $|B| \geq 3$. Thus $n \geq 3k + 3 \geq 3i + 6 > 2i + 6 = 2r$. If it preserves β' nontrivially, the complete internal-block lemma applies and gives the asserted terminal rank bound. It cannot fix β' pointwise: then $\beta \cup \beta'$ would be invariant under all of R, making it all of A and leaving the last generator trivial there. This accounts for every $d = 2$ possibility.

It remains to consider $d = 1$. Now $R = \langle \tau \rangle$, so $A = \beta \cup \beta'$ and $B = \{b, b'\}$. Thus $n = 2k + 2$ and $r = i + 2 \leq k + 1$. Failure of the rank bound would force $i = k - 1$ and $r = k + 1$. The involution τ exchanges the two k-point orbits and also exchanges b, b'; its evenness forces k odd, and $n \geq 12$ gives $k \geq 5$.

The faithful degree-k action of L has an independent string of length $k-1$. Whiston and the paid maximal symmetric-string classification identify it as the ordinary path marking of S_k; the degree-five certificate covers $k = 5$, and the uniform result covers odd $k \geq 7$. The subgroup $H = G_{<i-1}$ fixes a and $a' = a^\tau$ and acts diagonally as the natural S_{k-1} on the remaining $k-1$ pairs, one point from each of β, β'. These are its only two fixed points in A.

Write the crossing involution as $s = \rho_i = (a\ b)u$. The permutation u is supported on $A \setminus \{a\}$: the cross-edge is unique, and the other point b' of B cannot be moved. The group H fixes a, b and commutes with s, so u centralizes H. It fixes a, hence fixes the other H-fixed point a' as well. On the remaining $2(k-1)$ points the paired-row centralizer lemma applies, since $k-1$ is even and at least four. Thus u is even, making s odd, a contradiction. This completes the last case. □

Corollary 4 (Complete numerical support alternative)**.** *Let h be the largest index acting nontrivially on both A and B. Either $r \leq \lfloor (n-1)/2 \rfloor$, or $h < r - 1$ and, writing $X = \mathrm{supp}(G_{>h})$, one has*

$$X \subseteq B \setminus \{b\}, \qquad n - |X| \geq 2h + 1.$$

The latter inequality is strict if $h \geq i + 2$. At equality, $h = i + 1$, $k = i + 1$, $A = \beta \cup \beta'$, the fixed set is exactly $A \cup \{b\}$, and the faithful action of L on either of its two orbits is S_k.

Proof. The index h exists because τ moves both A and B. If the rank bound fails, Proposition 3 excludes $h = r - 1$. Every later generator is nonidentity and acts on exactly one of the two orbits. Connectedness of the string diagram forces all these generators to act on the same orbit: a change of sides would make an adjacent pair commute and disconnect the diagram. Proposition 2 excludes A as that side. Thus the tail fixes A pointwise, and the accepted orbit-distance support proposition gives every remaining assertion, including the strictness and equality parameters. □

Source, paid inputs and precise scope. This completes the numerical support alternative in the intransitive-left-side branch of Cameron–Fernandes–Leemans–Mixer, *Highest rank of a polytope for* A_n, arXiv:1605.09173v1, Proposition 5.23, physical pages 32–33. The source's commuting-block framework is retained; the orbit distance and tight-case parity arguments above are the present replacement proof. The exact fixed-set conclusion does not assert the source's complete coloured equality graph. All unique-crossing, active-orbit, internal-block and paired-row lemmas remain paid dependencies, as do the full L047 maximal symmetric-string theorem, its complete small certificate and the retained Whiston bounds. The internal-block input is its explicit nontrivial-B v2; the earlier frozen v1 is preserved historically. No new finite computation or formal declaration is needed. Singleton B, the residual small attachment and full-IP equality rigidity, later source tables, and the whole L044 contract remain separate. There is no new original, question or priority claim.